

Documentation Article - Salaries 5 Years post-grad

```
library(tidyverse)
library(vroom)
library(lmtest)
library(broom)
```

DATA

Salary -> Graduate's annual salary 5 years after graduation

MajorCategory -> Graduate's primary major listed on diploma

Sex -> Graduate's sex (M/F)

GPA -> Graduate's overall grade point average

```
sal <- vroom("Salary.csv", col_types = "dffd")
head(sal)
```

```
# A tibble: 6 x 4
  Salary MajorCategory Sex    GPA
  <dbl> <fct>          <fct> <dbl>
1 130000 Engineering      M     3.77
2  84200 Communications & Journalism F     2.79
3  58000 Education        F     3.79
4  95500 Business         F     3.9
5  68500 Communications & Journalism F     3.23
6 101000 Computers & Mathematics F     3.72
```

```
sal_lm <- lm(Salary ~ GPA + MajorCategory + Sex, data = sal)
summary(sal_lm)
```

Call:

```
lm(formula = Salary ~ GPA + MajorCategory + Sex, data = sal)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-16878.9	-3813.6	29.7	3824.1	19758.0

Coefficients:

	Estimate	Std. Error	t value
(Intercept)	99658.9	1697.5	58.71
GPA	5398.1	358.9	15.04
MajorCategoryCommunications & Journalism	-32602.5	1311.8	-24.85
MajorCategoryEducation	-41629.5	1310.2	-31.77
MajorCategoryBusiness	-21119.8	1242.4	-17.00
MajorCategoryComputers & Mathematics	-15637.6	1465.4	-10.67
MajorCategoryHealth	-23556.3	1387.8	-16.97
MajorCategoryArts	-42645.3	1470.8	-28.99
MajorCategorySocial Science	-29964.7	1337.6	-22.40
MajorCategoryLaw & Public Policy	-30854.4	1502.0	-20.54
MajorCategoryPhysical Sciences	-17405.3	1605.7	-10.84
MajorCategoryAgriculture & Natural Resources	-26519.0	1953.0	-13.58
MajorCategoryIndustrial Arts & Consumer Services	-30611.8	1624.0	-18.85
MajorCategoryPsychology & Social Work	-39668.2	1463.5	-27.11
MajorCategoryHumanities & Liberal Arts	-37975.8	1439.6	-26.38
MajorCategoryInterdisciplinary	-40989.1	2063.4	-19.86
MajorCategoryBiology & Life Science	-33953.8	1743.7	-19.47
SexF	-5659.2	420.7	-13.45

Pr(>|t|)

(Intercept)	<2e-16 ***
GPA	<2e-16 ***
MajorCategoryCommunications & Journalism	<2e-16 ***
MajorCategoryEducation	<2e-16 ***
MajorCategoryBusiness	<2e-16 ***
MajorCategoryComputers & Mathematics	<2e-16 ***
MajorCategoryHealth	<2e-16 ***
MajorCategoryArts	<2e-16 ***
MajorCategorySocial Science	<2e-16 ***
MajorCategoryLaw & Public Policy	<2e-16 ***
MajorCategoryPhysical Sciences	<2e-16 ***
MajorCategoryAgriculture & Natural Resources	<2e-16 ***
MajorCategoryIndustrial Arts & Consumer Services	<2e-16 ***
MajorCategoryPsychology & Social Work	<2e-16 ***

```
MajorCategoryHumanities & Liberal Arts      <2e-16 ***
MajorCategoryInterdisciplinary                <2e-16 ***
MajorCategoryBiology & Life Science          <2e-16 ***
SexF                                          <2e-16 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 5667 on 755 degrees of freedom
Multiple R-squared:  0.7854,    Adjusted R-squared:  0.7806
F-statistic: 162.6 on 17 and 755 DF,  p-value: < 2.2e-16
```

```
sal_inter_lm <- lm(Salary ~ GPA + MajorCategory + Sex + GPA:MajorCategory, data = sal)
summary(sal_inter_lm)
```

Call:

```
lm(formula = Salary ~ GPA + MajorCategory + Sex + GPA:MajorCategory,
    data = sal)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-16643.4	-3814.9	43.1	3771.3	19310.7

Coefficients:

	Estimate	Std. Error
(Intercept)	96194.2	8573.1
GPA	6404.7	2493.8
MajorCategoryCommunications & Journalism	-30560.1	9233.9
MajorCategoryEducation	-34935.0	9199.9
MajorCategoryBusiness	-21886.7	8914.2
MajorCategoryComputers & Mathematics	-1120.8	9825.6
MajorCategoryHealth	-19282.4	9688.8
MajorCategoryArts	-32097.1	10516.8
MajorCategorySocial Science	-31354.3	9284.8
MajorCategoryLaw & Public Policy	-32272.7	10563.2
MajorCategoryPhysical Sciences	-15460.1	12804.0
MajorCategoryAgriculture & Natural Resources	7656.9	17408.9
MajorCategoryIndustrial Arts & Consumer Services	-40989.4	10674.0
MajorCategoryPsychology & Social Work	-38153.3	10620.5
MajorCategoryHumanities & Liberal Arts	-26448.6	9647.3
MajorCategoryInterdisciplinary	-18590.2	15713.6
MajorCategoryBiology & Life Science	-25451.9	11657.6

SexF	-5596.9	422.4
GPA:MajorCategoryCommunications & Journalism	-594.8	2694.7
GPA:MajorCategoryEducation	-1990.9	2686.7
GPA:MajorCategoryBusiness	275.2	2603.2
GPA:MajorCategoryComputers & Mathematics	-4450.7	2896.1
GPA:MajorCategoryHealth	-1270.5	2855.2
GPA:MajorCategoryArts	-3110.7	3067.9
GPA:MajorCategorySocial Science	478.7	2721.9
GPA:MajorCategoryLaw & Public Policy	422.3	3080.2
GPA:MajorCategoryPhysical Sciences	-561.8	3771.0
GPA:MajorCategoryAgriculture & Natural Resources	-9678.5	4931.0
GPA:MajorCategoryIndustrial Arts & Consumer Services	3241.7	3152.6
GPA:MajorCategoryPsychology & Social Work	-433.2	3116.1
GPA:MajorCategoryHumanities & Liberal Arts	-3478.6	2827.6
GPA:MajorCategoryInterdisciplinary	-6602.2	4590.6
GPA:MajorCategoryBiology & Life Science	-2597.6	3496.7
	t value	Pr(> t)
(Intercept)	11.220	< 2e-16 ***
GPA	2.568	0.010415 *
MajorCategoryCommunications & Journalism	-3.310	0.000980 ***
MajorCategoryEducation	-3.797	0.000158 ***
MajorCategoryBusiness	-2.455	0.014307 *
MajorCategoryComputers & Mathematics	-0.114	0.909217
MajorCategoryHealth	-1.990	0.046939 *
MajorCategoryArts	-3.052	0.002355 **
MajorCategorySocial Science	-3.377	0.000771 ***
MajorCategoryLaw & Public Policy	-3.055	0.002330 **
MajorCategoryPhysical Sciences	-1.207	0.227646
MajorCategoryAgriculture & Natural Resources	0.440	0.660190
MajorCategoryIndustrial Arts & Consumer Services	-3.840	0.000134 ***
MajorCategoryPsychology & Social Work	-3.592	0.000349 ***
MajorCategoryHumanities & Liberal Arts	-2.742	0.006262 **
MajorCategoryInterdisciplinary	-1.183	0.237163
MajorCategoryBiology & Life Science	-2.183	0.029328 *
SexF	-13.251	< 2e-16 ***
GPA:MajorCategoryCommunications & Journalism	-0.221	0.825356
GPA:MajorCategoryEducation	-0.741	0.458919
GPA:MajorCategoryBusiness	0.106	0.915834
GPA:MajorCategoryComputers & Mathematics	-1.537	0.124773
GPA:MajorCategoryHealth	-0.445	0.656471
GPA:MajorCategoryArts	-1.014	0.310944
GPA:MajorCategorySocial Science	0.176	0.860448
GPA:MajorCategoryLaw & Public Policy	0.137	0.890992

GPA:MajorCategoryPhysical Sciences	-0.149	0.881605
GPA:MajorCategoryAgriculture & Natural Resources	-1.963	0.050048
GPA:MajorCategoryIndustrial Arts & Consumer Services	1.028	0.304169
GPA:MajorCategoryPsychology & Social Work	-0.139	0.889480
GPA:MajorCategoryHumanities & Liberal Arts	-1.230	0.219008
GPA:MajorCategoryInterdisciplinary	-1.438	0.150800
GPA:MajorCategoryBiology & Life Science	-0.743	0.457792

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

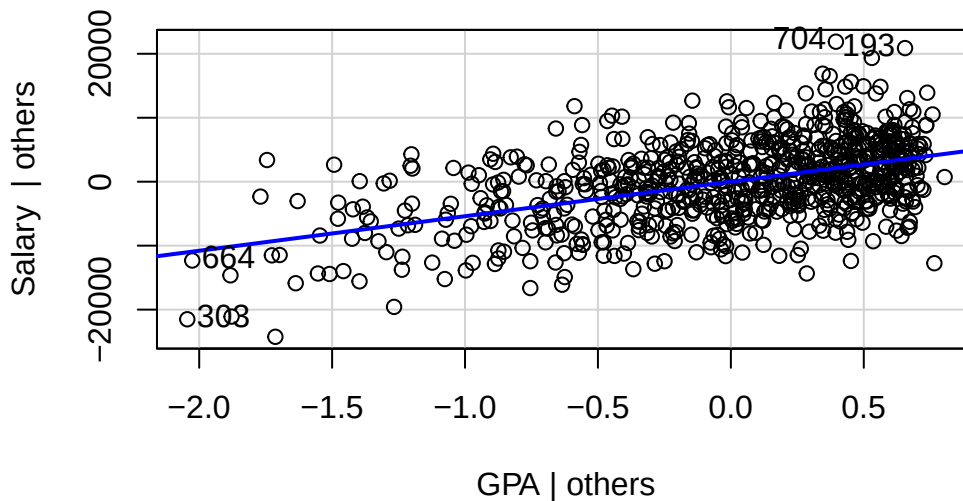
Residual standard error: 5616 on 740 degrees of freedom

Multiple R-squared: 0.7935, Adjusted R-squared: 0.7845

F-statistic: 88.84 on 32 and 740 DF, p-value: < 2.2e-16

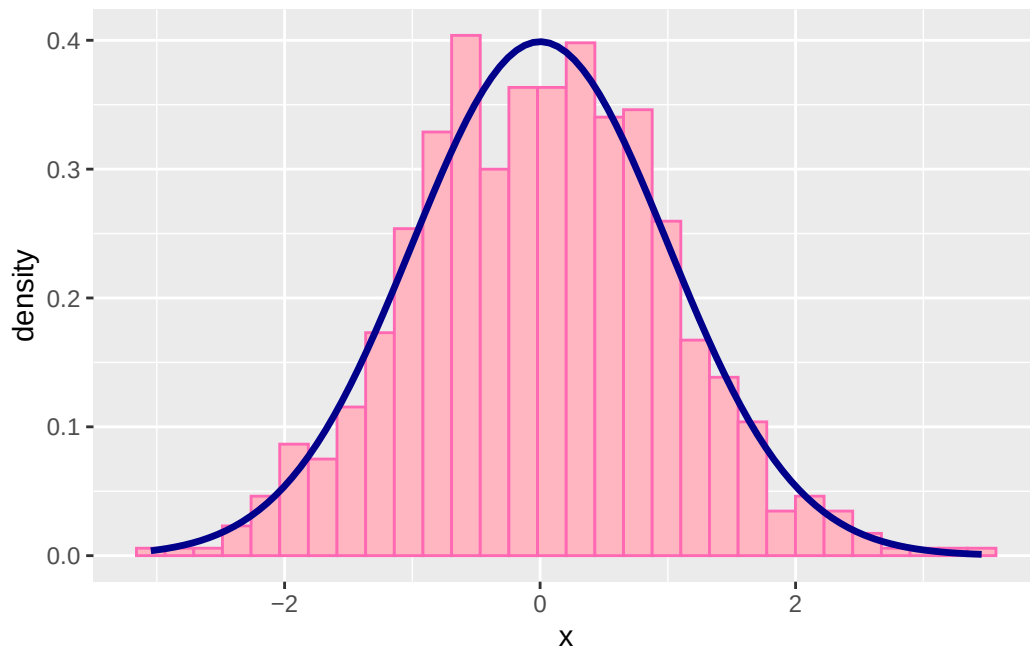
Linearity

```
# eda
# linearity added variable plot
car::avPlots(sal_lm, terms=~GPA, ask=FALSE)
```



```
# normality (hist of residuals and KS test)
std_resids <- data.frame(x=MASS::stdres(sal_inter_lm))
```

```
ggplot(data = std_resids,
       mapping = aes(x = x)) +
  geom_histogram(aes(y = after_stat(density)),
                bins = 30,
                col = "hotpink",
                fill = "lightpink") +
  stat_function(fun = dnorm,
               args = list(mean = 0, sd = 1),
               color = "darkblue",
               linewidth = 1.2)
```



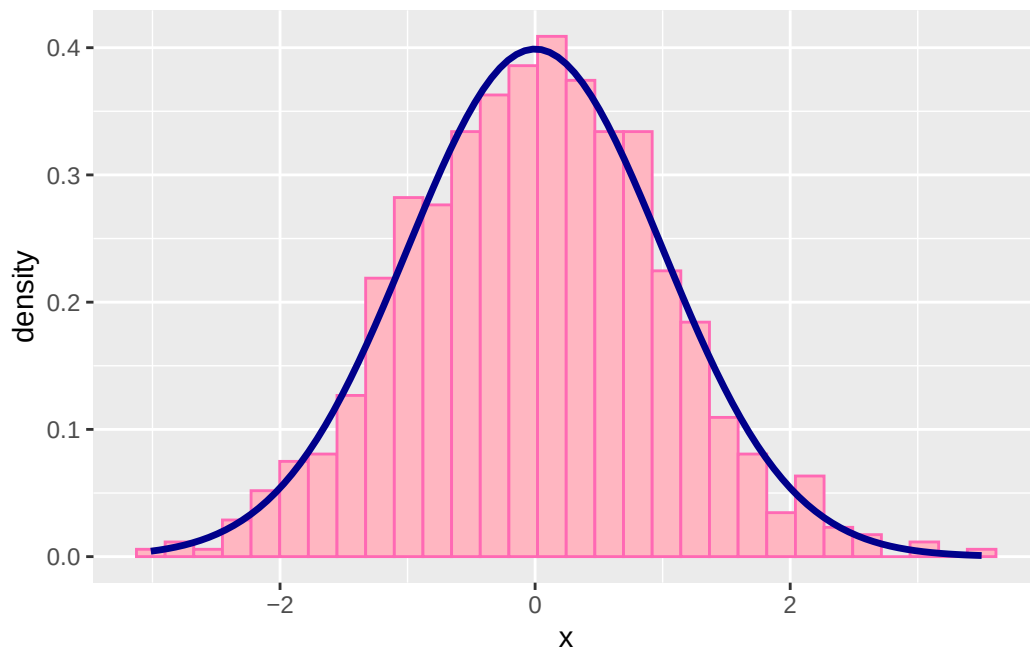
```
ks.test(x = std_resids, y = "pnorm", mean = 0, sd = 1)
```

Warning in ks.test.default(x = std_resids, y = "pnorm", mean = 0, sd = 1): ties should not be present for the one-sample Kolmogorov-Smirnov test

Asymptotic one-sample Kolmogorov-Smirnov test

```
data:  std_resids
D = 0.02187, p-value = 0.8535
alternative hypothesis: two-sided
```

```
std_resids1 <- data.frame(x=MASS::stdres(sal_lm))
ggplot(data = std_resids1,
       mapping = aes(x = x)) +
  geom_histogram(aes(y = after_stat(density)),
                bins = 30,
                col = "hotpink",
                fill = "lightpink") +
  stat_function(fun = dnorm,
               args = list(mean = 0, sd = 1),
               color = "darkblue",
               linewidth = 1.2)
```



```
ks.test(x = std_resids1, y = "pnorm", mean = 0, sd = 1)
```

Warning in ks.test.default(x = std_resids1, y = "pnorm", mean = 0, sd = 1):
ties should not be present for the one-sample Kolmogorov-Smirnov test

Asymptotic one-sample Kolmogorov-Smirnov test

data: std_resids1

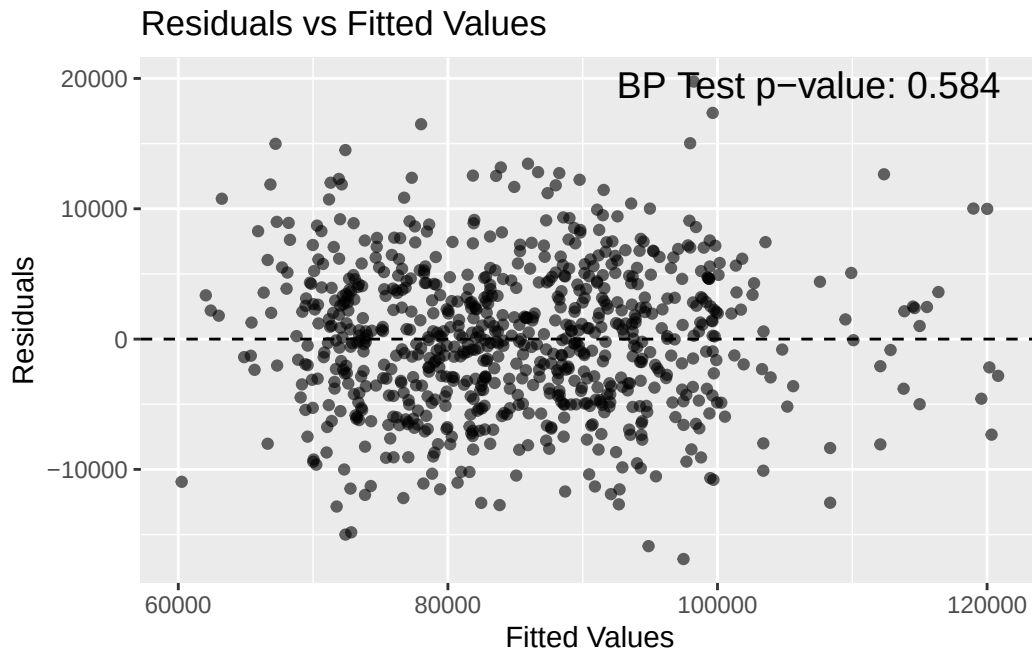
D = 0.016463, p-value = 0.9848
alternative hypothesis: two-sided

```
# equal variance (fitted vs resid and bp test)
# get fitted values and residuals
fitted_vals <- fitted(sal_lm)
resids <- resid(sal_lm)

# run BP test
bp_test <- bptest(sal_lm)

# extract p-value and format to 3 decimals
bp_pval <- round(bp_test$p.value, 3)

# plot: residuals vs fitted with BP p-value annotation
ggplot(data.frame(fitted = fitted_vals, resid = resids),
  aes(x = fitted, y = resid)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = "Fitted Values",
    y = "Residuals",
    title = "Residuals vs Fitted Values") +
  annotate("text",
    x = Inf,
    y = Inf,
    label = paste("BP Test p-value:", bp_pval),
    hjust = 1.1,
    vjust = 1.5,
    size = 5)
```

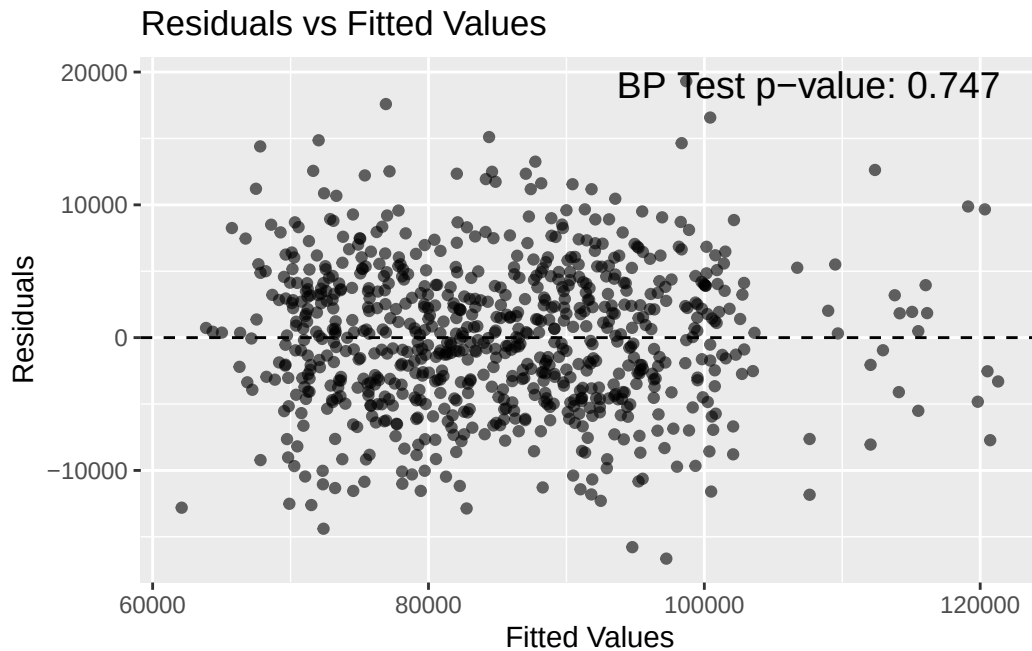
```
# equal variance (fitted vs resid and bp test)
# get fitted values and residuals
fitted_vals <- fitted(sal_inter_lm)
resids <- resid(sal_inter_lm)

# run BP test
bp_test <- bptest(sal_inter_lm)

# extract p-value and format to 3 decimals
bp_pval <- round(bp_test$p.value, 3)

# plot: residuals vs fitted with BP p-value annotation
ggplot(data.frame(fitted = fitted_vals, resid = resids),
  aes(x = fitted, y = resid)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = "Fitted Values",
    y = "Residuals",
    title = "Residuals vs Fitted Values") +
  annotate("text",
    x = Inf,
    y = Inf,
    label = paste("BP Test p-value:", bp_pval),
```

```
hjust = 1.1,
vjust = 1.5,
size = 5)
```



Independence: yes, each student is independent from one another.

2. Impact of GPA → What impact, if any, does GPA have on salary? Do you think students should be hyper-anxious about grades? Is the effect of GPA different for different majors (does GPA matter more for some majors than others)?

```
tidy(sal_lm) |> filter(term == "GPA")
```

```
# A tibble: 1 x 5
  term estimate std.error statistic p.value
<chr>   <dbl>     <dbl>     <dbl>   <dbl>
1 GPA     5398.       359.      15.0 6.52e-45
```

```
tidy(sal_inter_lm) |> filter(grepl("GPA", term))
```

```
# A tibble: 16 x 5
```

	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	GPA	6405.	2494.	2.57	0.0104
2	GPA:MajorCategoryCommunications & Journ~	-595.	2695.	-0.221	0.825
3	GPA:MajorCategoryEducation	-1991.	2687.	-0.741	0.459
4	GPA:MajorCategoryBusiness	275.	2603.	0.106	0.916
5	GPA:MajorCategoryComputers & Mathematics	-4451.	2896.	-1.54	0.125
6	GPA:MajorCategoryHealth	-1270.	2855.	-0.445	0.656
7	GPA:MajorCategoryArts	-3111.	3068.	-1.01	0.311
8	GPA:MajorCategorySocial Science	479.	2722.	0.176	0.860
9	GPA:MajorCategoryLaw & Public Policy	422.	3080.	0.137	0.891
10	GPA:MajorCategoryPhysical Sciences	-562.	3771.	-0.149	0.882
11	GPA:MajorCategoryAgriculture & Natural ~	-9678.	4931.	-1.96	0.0500
12	GPA:MajorCategoryIndustrial Arts & Cons~	3242.	3153.	1.03	0.304
13	GPA:MajorCategoryPsychology & Social Wo~	-433.	3116.	-0.139	0.889
14	GPA:MajorCategoryHumanities & Liberal A~	-3479.	2828.	-1.23	0.219
15	GPA:MajorCategoryInterdisciplinary	-6602.	4591.	-1.44	0.151
16	GPA:MajorCategoryBiology & Life Science	-2598.	3497.	-0.743	0.458

GPA does have a statistically significant impact on Salary, potentially affecting thousands of dollars per GPA point.

3. Salaries Across Majors → Do you see differences in salaries across majors? If so, what majors seem to make more money?

```
tidy(sal_lm) |> filter(grepl("MajorCategory", term))
```

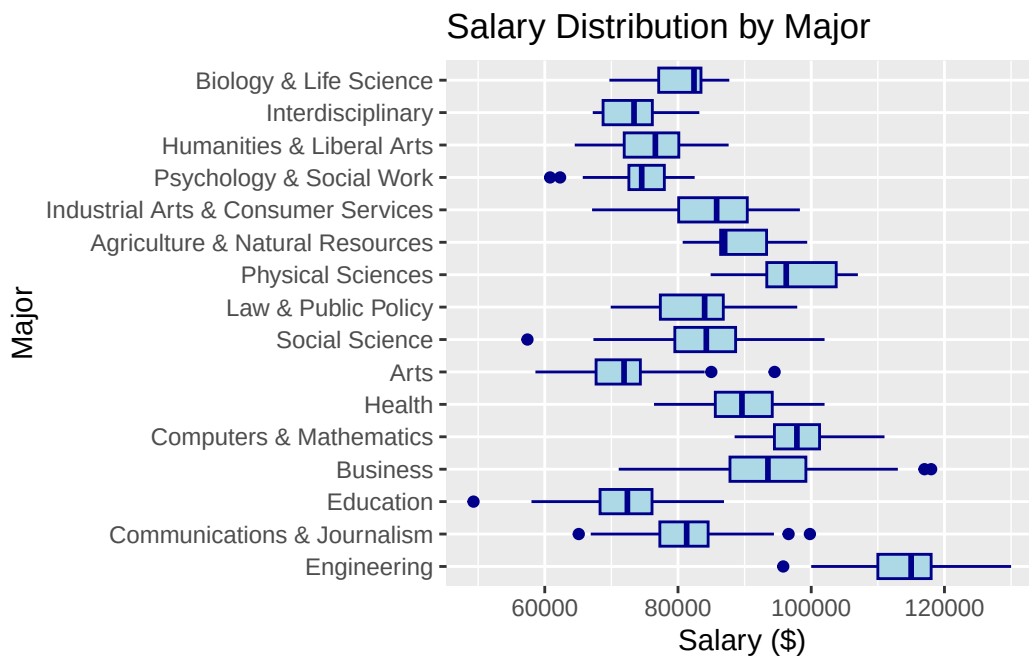
```
# A tibble: 15 x 5
```

	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	MajorCategoryCommunications & Journal~	-32603.	1312.	-24.9	4.29e-100
2	MajorCategoryEducation	-41630.	1310.	-31.8	2.54e-141
3	MajorCategoryBusiness	-21120.	1242.	-17.0	4.10e- 55
4	MajorCategoryComputers & Mathematics	-15638.	1465.	-10.7	7.37e- 25
5	MajorCategoryHealth	-23556.	1388.	-17.0	5.58e- 55
6	MajorCategoryArts	-42645.	1471.	-29.0	8.25e-125
7	MajorCategorySocial Science	-29965.	1338.	-22.4	1.28e- 85
8	MajorCategoryLaw & Public Policy	-30854.	1502.	-20.5	7.82e- 75
9	MajorCategoryPhysical Sciences	-17405.	1606.	-10.8	1.52e- 25

10	MajorCategoryAgriculture & Natural Re~	-26519.	1953.	-13.6	9.79e- 38
11	MajorCategoryIndustrial Arts & Consum~	-30612.	1624.	-18.8	3.03e- 65
12	MajorCategoryPsychology & Social Work	-39668.	1463.	-27.1	1.56e-113
13	MajorCategoryHumanities & Liberal Arts	-37976.	1440.	-26.4	3.36e-109
14	MajorCategoryInterdisciplinary	-40989.	2063.	-19.9	5.75e- 71
15	MajorCategoryBiology & Life Science	-33954.	1744.	-19.5	9.57e- 69

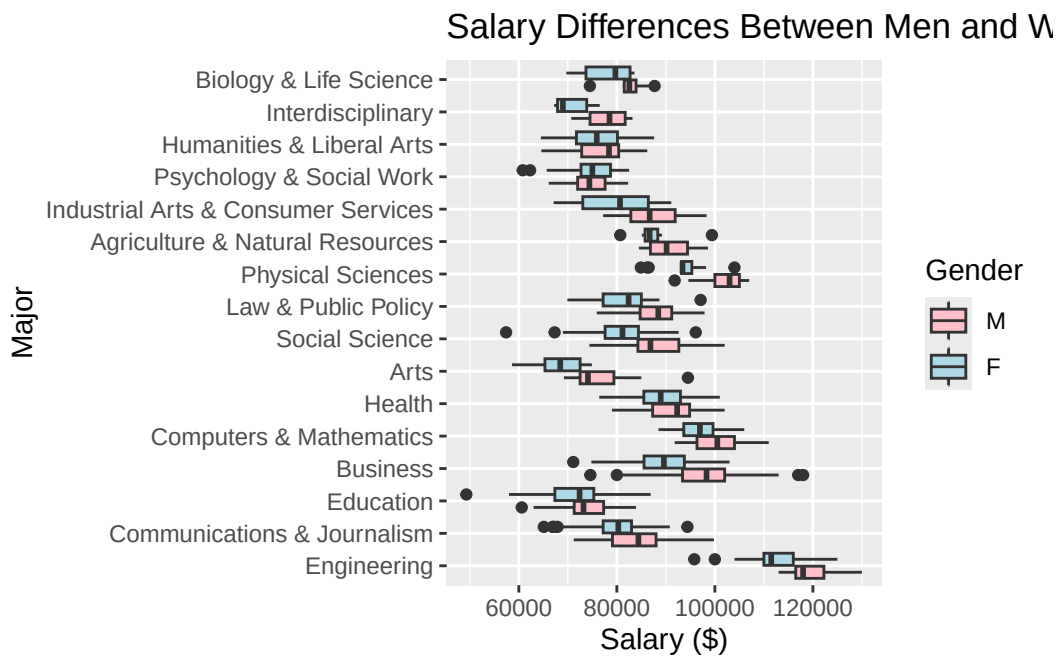
boxplot code for across majors

```
# Boxplot of salaries by major
ggplot(sal, aes(x = MajorCategory, y = Salary)) +
  geom_boxplot(fill = "lightblue", color = "darkblue") +
  coord_flip() +
  labs(title = "Salary Distribution by Major",
       x = "Major",
       y = "Salary ($)")
```



4. Salaries for Women and Men → Are there differences in salaries for men vs. women within the same major category? If so, does the difference in salary between men vs. women differ by major category? If there are differences, demonstrate what the difference is between men and women salary for your major category as an example.

```
ggplot(sal, aes(x = MajorCategory, y = Salary, fill = Sex)) +
  geom_boxplot(position = position_dodge(width = 0.8)) +
  coord_flip() +
  labs(title = "Salary Differences Between Men and Women by Major",
       x = "Major",
       y = "Salary ($)",
       fill = "Gender") +
  scale_fill_manual(values = c("pink", "lightblue"))
```



```
sal %>%
  group_by(MajorCategory, Sex) %>%
  summarize(mean_salary = mean(Salary, na.rm = TRUE)) %>%
  pivot_wider(names_from = Sex, values_from = mean_salary) %>%
  mutate(diff = M - F) %>%
  arrange(desc(abs(diff)))
```

`summarise()` has grouped output by 'MajorCategory'. You can override using the `groups` argument.

```
# A tibble: 16 x 4
# Groups:   MajorCategory [16]
  MajorCategory      M      F  diff
  <fct>          <dbl> <dbl> <dbl>
1 Engineering    120000 111362. 8638.
2 Arts           76735. 68491. 8244.
3 Physical Sciences 101523. 93446. 8077.
4 Business       97414. 89420. 7994.
5 Industrial Arts & Consumer Services 87327. 79860 7467.
6 Social Science  88153. 80795 7358.
7 Interdisciplinary 77725 70886. 6839.
8 Law & Public Policy 87580 81224 6356
9 Biology & Life Science 82630 77533. 5097.
10 Communications & Journalism 84372 79783. 4589.
11 Computers & Mathematics 100450 96514. 3936.
12 Agriculture & Natural Resources 90833. 87857. 2976.
13 Education      74015. 71231. 2784.
14 Health         91500 88889. 2611.
15 Humanities & Liberal Arts 76940 75725 1215
16 Psychology & Social Work 74367. 74484 -117.
```

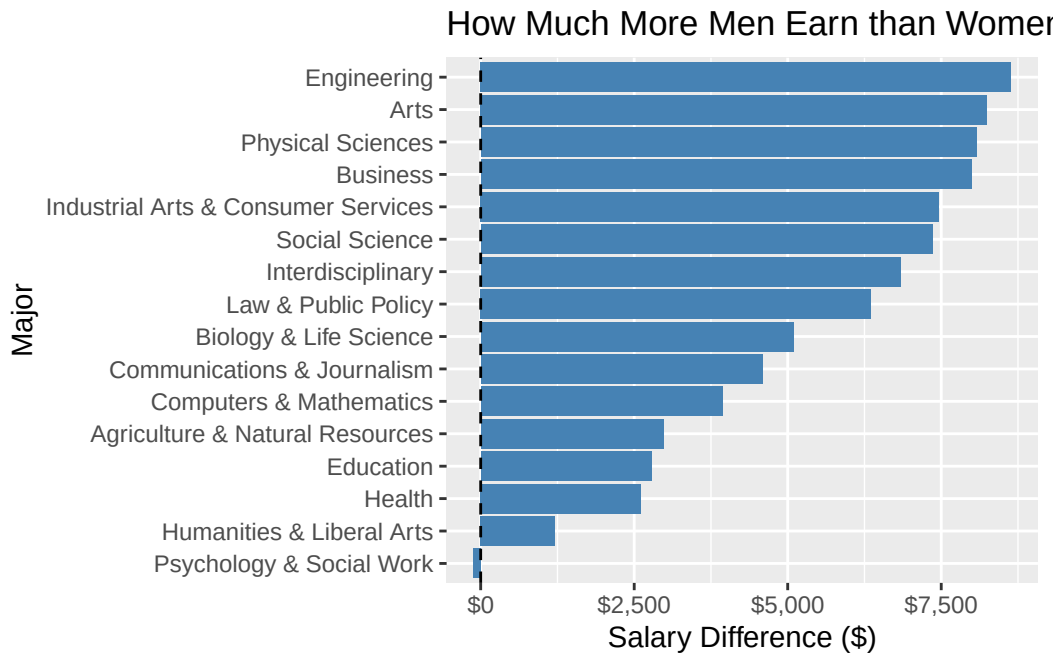
```
major_salary <- sal %>%
  group_by(MajorCategory) %>%
  summarize(
    mean_salary = mean(Salary, na.rm = TRUE),
    .groups = "drop"
  )
```

```
sal %>%
  group_by(MajorCategory, Sex) %>%
  summarize(mean_salary = mean(Salary, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(names_from = Sex, values_from = mean_salary) %>%
  mutate(diff = M - F) %>%
  ggplot(aes(x = reorder(MajorCategory, diff), y = diff)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(
    title = "How Much More Men Earn than Women by Major",
```

```

x = "Major",
y = "Salary Difference ($)"
) +
scale_y_continuous(labels = scales::dollar_format())

```

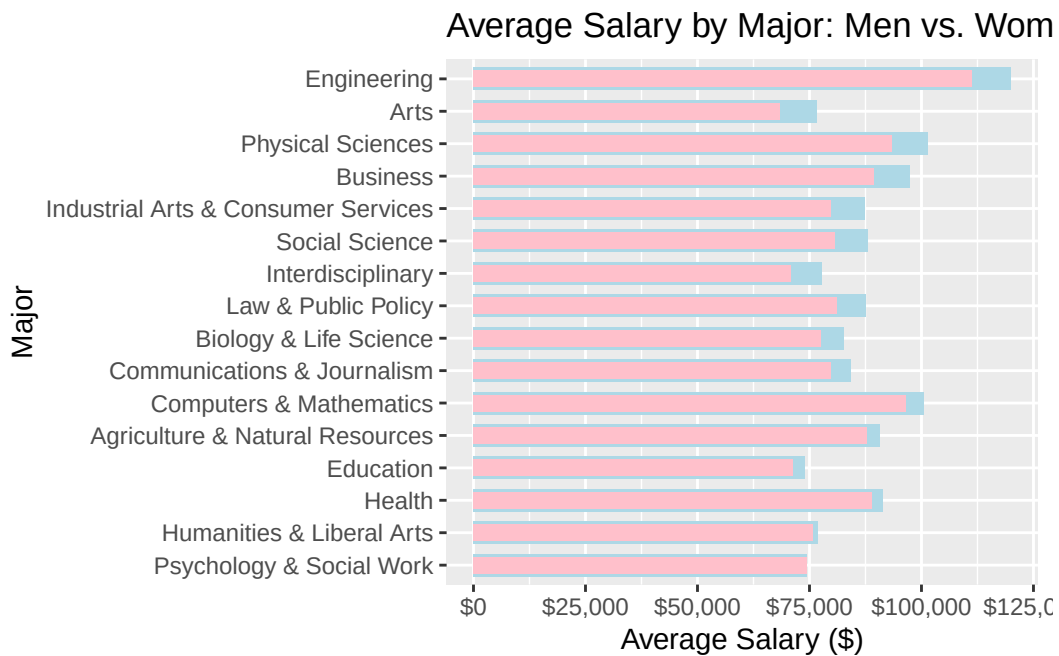


```

avg_sal <- sal %>%
  group_by(MajorCategory, Sex) %>%
  summarize(mean_salary = mean(Salary, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(names_from = Sex, values_from = mean_salary)
avg_sal <- avg_sal %>%
  mutate(diff = M - F) %>%
  arrange(diff) %>%
  mutate(MajorCategory = factor(MajorCategory, levels = MajorCategory))
ggplot(avg_sal, aes(x = MajorCategory)) +
  geom_col(aes(y = M), fill = "lightblue", width = 0.7) +
  geom_col(aes(y = F), fill = "pink", width = 0.5) +
  coord_flip() +
  labs(
    title = "Average Salary by Major: Men vs. Women",
    x = "Major",
    y = "Average Salary ($)"
  ) +

```

```
scale_y_continuous(labels = scales::dollar_format())
```



```
cm_data <- sal %>%
  filter(MajorCategory == "Computers & Mathematics")
t.test(
  Salary ~ Sex,
  data = cm_data,
  conf.level = 0.95
)
```

Welch Two Sample t-test

data: Salary by Sex

t = 2.3554, df = 34.777, p-value = 0.02427

alternative hypothesis: true difference in means between group M and group F is not equal to 0
95 percent confidence interval:

542.8894 7329.8379

sample estimates:

mean in group M mean in group F

100450.00 96513.64

5. Other Factors –> How well do the variables you have in your dataset determine salary? Do you think that there are other variables that contribute to salary? Use a specific example to demonstrate how well (or not well) the factors can be used to determine (i.e. predict) salary.